Exercise 2 — Step-by-Step Solutions

Advanced System Theory · System Properties and Linearization

Course: Advanced System Theory (Paderborn University, Automatic Control Group RAT, Prof. Dr. Erdal Kayacan) **Covers:** Task 1 (a–d), Task 2 (a, b and the I/O-relation part, called "2c" here because the sheet labels it "b" twice), Task 3 (a–d) **Checking:** no official solution was available when this was written. Every result is verified by the scripts in `python/` (symbolic algebra with SymPy, numerical search, simulation).

Every step is written out. References like "Concepts §3.2" point to `concepts.md`.

Contents

- Task 1 — System properties
- Task 2a — Linearization of $\dot{x}_1 = -\sin x_1,\ \dot{x}_2 = -x_2 + \cos x_1$
- Task 2b — Linearization of the second nonlinear system
- Task 2c — State space and linearization of the I/O relation
- Task 3 — Evolution of a discrete-time system
- Summary of results

Task 1 — System properties

For each system we test **causality** (Concepts §4.2), **time invariance** (two-path test, §3.2) and **linearity** (§2.3). A "Yes" is proved in general; a "No" is shown with one counterexample (§1.3). Throughout, $A \neq 0$ and $f_0 \neq 0$ (special cases are noted at the end).

1(a) $y(t) = u(t) - 0.5\,u(t+1)$

Causality. The output at time t reads the input at t and at $t+1$. The read time $t+1 > t$ lies in the future for every t.

Counterexample. Fix any T. Let $u_1 \equiv 0$ and $u_2(t) = 0$ for $t \leq T$, $u_2(t) = 1$ for $t > T$. The inputs agree for all $t \leq T$. At $t = T$: $y_1(T) = 0$, but $y_2(T) = u_2(T) - 0.5\,u_2(T+1) = 0 - 0.5 = -0.5$. The outputs differ at a time $\leq T$ ⇒ **not causal**.

Time invariance. - Path 1 (shift, then system): input $u_T(t) = u(t-T)$ gives $y_T(t) = u_T(t) - 0.5\,u_T(t+1) = u(t-T) - 0.5\,u(t+1-T)$. - Path 2 (system, then shift): replace every t by $t-T$: $y(t-T) = u(t-T) - 0.5\,u(t-T+1)$. - Identical for every u, t, T ⇒ **time-invariant**.

Linearity. Let $u_3 = \alpha_1 u_1 + \alpha_2 u_2$:

$$y_3(t) = u_3(t) - 0.5\,u_3(t+1) = \alpha_1 u_1(t) + \alpha_2 u_2(t) - 0.5\big(\alpha_1 u_1(t+1) + \alpha_2 u_2(t+1)\big) = \alpha_1 \underbrace{\big(u_1(t) - 0.5 u_1(t+1}_{y_1(t)}$$

⇒ **linear**.

Interpretation: an LTI system that "looks one second ahead" — perfectly linear and time-invariant, but cannot be built as a real-time device.

1(b) $y(t) = u(2t)$

Causality. The read time is $s(t) = 2t$. For $t > 0$, $2t > t$: the output needs future input values. (For $t < 0$ it reads the past, but causality must hold for all t.)

Counterexample. $T = 1$, $u_1 \equiv 0$, $u_2(t) = 0$ for $t \leq 1$ and 1 for $t > 1$. They agree for $t \leq 1$. At $t = 0.75 \leq 1$: $y_1(0.75) = u_1(1.5) = 0$, $y_2(0.75) = u_2(1.5) = 1$ ⇒ **not causal**.

Time invariance. Careful use of the "replace t" rule (Concepts §1.2): - Path 1: $u_T(t) = u(t-T)$. The system reads its input at time $2t$, so $y_T(t) = u_T(2t) = u(2t - T)$. - Path 2: $y(t - T) = u\big(2(t - T)\big) = u(2t - 2T)$. - These differ: the delay at the output is $T/2$ instead of T (the compression also compresses the delay).

Counterexample. u = unit step ($u(t) = 1$ for $t \geq 0$, else 0), $T = 1$, $t = 0.75$: path 1 gives $u(0.5) = 1$, path 2 gives $u(-0.5) = 0$ ⇒ **time-variant**.

Linearity. With $u_3 = \alpha_1 u_1 + \alpha_2 u_2$: $y_3(t) = u_3(2t) = \alpha_1 u_1(2t) + \alpha_2 u_2(2t) = \alpha_1 y_1(t) + \alpha_2 y_2(t)$ ⇒ **linear**. (Changing the time argument never destroys linearity.)

Interpretation: playing a recording at double speed. It is linear, but a 1 s delay of the recording shows up as only 0.5 s at the output, and to play at double speed in real time you would need the future of the recording.

1(c) $y(t) = A\cos\big(2\pi f_0 t + u(t)\big)$

This is **phase modulation**: the input sets the phase of a carrier.

Causality. Memoryless: $y(t)$ uses u only at the same instant t ⇒ **causal**. (If $u_1 = u_2$ for $t \leq T$, then $y_1(t) = A\cos(2\pi f_0 t + u_1(t)) = A\cos(2\pi f_0 t + u_2(t)) = y_2(t)$ for $t \leq T$.)

Time invariance. - Path 1: $y_T(t) = A\cos\big(2\pi f_0 t + u(t - T)\big)$ — the carrier term $2\pi f_0 t$ is *not* shifted, because it is part of the system, not of the input. - Path 2: $y(t - T) = A\cos\big(2\pi f_0 (t - T) + u(t - T)\big)$.

Counterexample. $u \equiv 0$, $T = \frac{1}{4f_0}$ (a quarter period), $t = 0$: path 1 gives $A\cos 0 = A$; path 2 gives $A\cos(-\pi/2) = 0$. Different ⇒ **time-variant**. The explicit t in the carrier is the red flag (Concepts §3.3).

Linearity. Zero-in-zero-out test (§2.3): $u \equiv 0$ gives $y(t) = A\cos(2\pi f_0 t) \not\equiv 0$ ⇒ **not linear**. (Also homogeneity fails in general: $A\cos(2\pi f_0 t + 2u) \neq 2A\cos(2\pi f_0 t + u)$; the input sits inside the cosine.)

1(d) $y(t) = u(t)\cos(2\pi f_0 t + \phi)$

This is **amplitude modulation** (a mixer): the input multiplies a carrier.

Causality. Memoryless ⇒ **causal**.

Time invariance. - Path 1: $y_T(t) = u(t - T)\cos(2\pi f_0 t + \phi)$. - Path 2: $y(t - T) = u(t - T)\cos\big(2\pi f_0 (t - T) + \phi\big)$.

Counterexample. $u \equiv 1$, $T = \frac{1}{4f_0}$, $t = -\frac{\phi}{2\pi f_0}$ (so that $2\pi f_0 t + \phi = 0$): path 1 gives $\cos 0 = 1$, path 2 gives $\cos(-\pi/2) = 0$ ⇒ **time-variant**.

Linearity. Let $c(t) = \cos(2\pi f_0 t + \phi)$, a known function of time that does not depend on u:

$$y_3(t) = (\alpha_1 u_1(t) + \alpha_2 u_2(t))c(t) = \alpha_1 u_1(t)c(t) + \alpha_2 u_2(t)c(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t)$$

⇒ **linear** (a linear, time-varying system).

Compare (c) and (d): both contain a carrier with an explicit t, so both are time-variant. In (d) the input multiplies the cosine (linear); in (c) it sits inside the cosine (nonlinear).

Summary table and special cases

System	Causal	Time-invariant	Linear
a) $u(t) - 0.5\,u(t+1)$	N	Y	Y
b) $u(2t)$	N	N	Y
c) $A\cos(2\pi f_0 t + u(t))$	Y	N	N
d) $u(t)\cos(2\pi f_0 t + \phi)$	Y	N	Y

Special parameter values. If $f_0 = 0$, the explicit time dependence disappears: (c) becomes $A\cos(u(t))$ and (d) becomes $u(t)\cos\phi$, both time-invariant; (c) stays nonlinear unless $A = 0$ (then $y \equiv 0$, which is trivially linear and TI).

Extra (not asked): (a) and (d) are BIBO stable ($|y| \le 1.5M$ and $|y| \le M$); (b) and (c) too ($|y| \le M$, $|y| \le |A|$).

Check: `python/task1_system_properties.py` tests all three properties numerically on random signals and reproduces the table and both counterexamples.

Task 2a — Linearization

$$\dot{x}_1 = -\sin x_1 =: f_1(x), \qquad \dot{x}_2 = -x_2 + \cos x_1 =: f_2(x).$$

The system is autonomous (no input), so the linear model is $\Delta\dot{x} = A\,\Delta x$.

Step 1 — Equilibria

Solve $f_1 = 0$, $f_2 = 0$ (Concepts §8.2). The first equation contains only x_1:

$$-\sin x_1 = 0 \iff x_1 = k\pi, \quad k \in \mathbb{Z}.$$

Substitute into the second: $-x_2 + \cos(k\pi) = 0 \iff x_2 = \cos(k\pi) = (-1)^k$.

$$\boxed{x_0^{(k)} = (k\pi,\ (-1)^k), \qquad k \in \mathbb{Z}}$$

So there are infinitely many isolated equilibria, in two families: - k even: $(0,1), (\pm 2\pi, 1), (\pm 4\pi, 1), \ldots$ - k odd: $(\pm\pi, -1), (\pm 3\pi, -1), \ldots$

Check $k = 1$: $f_1 = -\sin\pi = 0$, $f_2 = -(-1) + \cos\pi = 1 - 1 = 0$ ✓.

Step 2 — Jacobian (symbolic)

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} -\cos x_1 & 0 \\ -\sin x_1 & -1 \end{bmatrix}.$$

($\partial(-\sin x_1)/\partial x_1 = -\cos x_1$; f_1 does not contain x_2; $\partial(\cos x_1)/\partial x_1 = -\sin x_1$; $\partial(-x_2)/\partial x_2 = -1$.)

Step 3 — Evaluate at the equilibria

Use $\cos(k\pi) = (-1)^k$, $\sin(k\pi) = 0$:

$$A^{(k)} = \begin{bmatrix} -(-1)^k & 0 \\ 0 & -1 \end{bmatrix} \quad \Longrightarrow \quad A_{\text{even}} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad A_{\text{odd}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Step 4 — Linearized models

With $\Delta x = x - x_0^{(k)}$:

$$\boxed{k \text{ even}: \begin{bmatrix} \Delta\dot{x}_1 \\ \Delta\dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix}, \qquad k \text{ odd}: \begin{bmatrix} \Delta\dot{x}_1 \\ \Delta\dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix}}$$

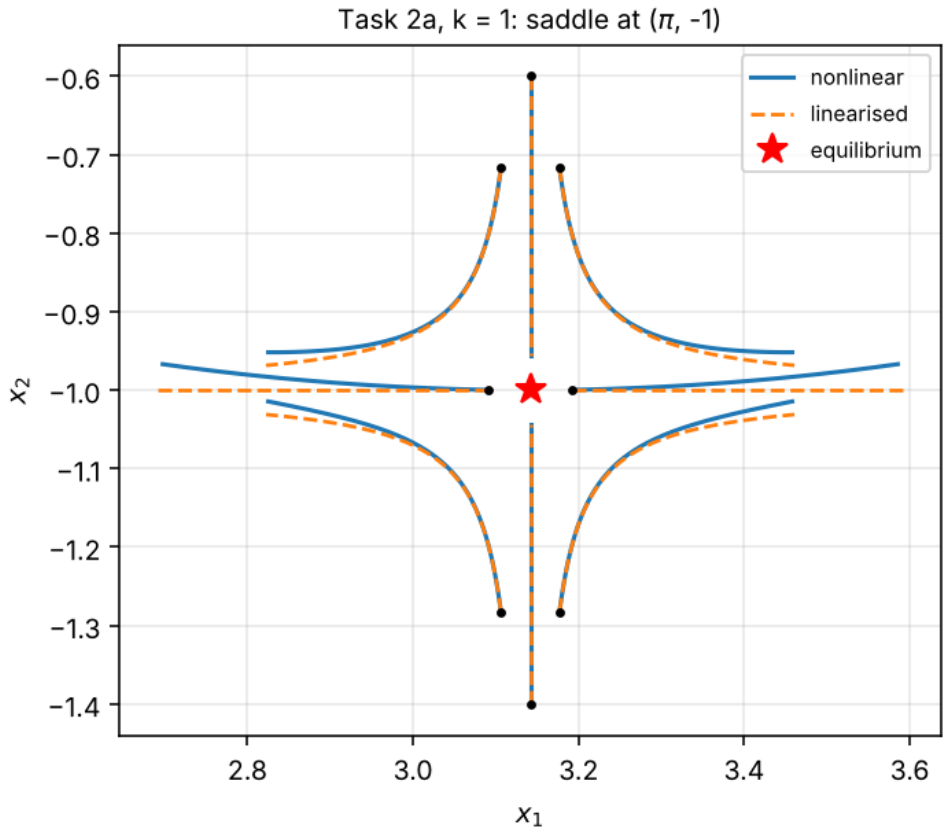
where, e.g. for $k = 0$, $\Delta x_1 = x_1$, $\Delta x_2 = x_2 - 1$; for $k = 1$, $\Delta x_1 = x_1 - \pi$, $\Delta x_2 = x_2 + 1$.

Step 5 — Interpretation (Concepts §10)

The matrices are diagonal, so the eigenvalues are the diagonal entries. - k even: $\lambda = -1, -1$ ⇒ both negative ⇒ **locally asymptotically stable** (stable node). Small deviations decay like e^{-t}. - k odd: $\lambda = +1, -1$ ⇒ **saddle, unstable**. Deviations in x_1 grow like e^{t}; deviations in x_2 decay.

Physically this is the same structure as the pendulum: $\dot{x}_1 = -\sin x_1$ pushes x_1 towards the nearest even multiple of π and away from odd multiples; x_2 simply follows $\cos x_1$ with a first-order lag.

Check: `python/task2_linearization.py` (symbolic Jacobians for $k = -2, \ldots, 2$); figures `task2a_k0_stable.png` and `task2a_k1_saddle.png` overlay nonlinear and linearized trajectories.



Task 2b — Linearization of the second nonlinear system

$$\dot{x}_1 = x_1^2 + x_2^2 + x_2 \cos x_1 =: f_1(x),$$
$$\dot{x}_2 = (1 + x_1)x_1 + (1 + x_2)x_2 + x_1 \sin x_2 = x_1 + x_1^2 + x_2 + x_2^2 + x_1 \sin x_2 =: f_2(x).$$

Step 1 — Equilibria

We need

$$\text{(I)} \quad x_1^2 + x_2^2 + x_2 \cos x_1 = 0, \qquad \text{(II)} \quad x_1^2 + x_2^2 + x_1 + x_2 + x_1 \sin x_2 = 0.$$

Obvious candidate. $x = (0, 0)$: (I) $= 0$, (II) $= 0$ ✓.

Look along $x_1 = 0$**.** Then (I) becomes $x_2^2 + x_2 = x_2(x_2 + 1) = 0$ and (II) becomes $x_2^2 + x_2 = 0$ — the same equation. So $x_2 = 0$ or $x_2 = -1$:

$$(0, 0) \quad \text{and} \quad (0, -1).$$

Check $(0, -1)$*:* (I) $= 0 + 1 + (-1)\cos 0 = 0$ ✓; (II) $= 0 + 1 + 0 - 1 + 0 = 0$ ✓.

Are there others? We show there are none (Concepts §8.2, steps 4–5).

(i) Confine the solutions. From (I), $x_1^2 + x_2^2 = -x_2 \cos x_1 \le |x_2|$ (because $|\cos| \le 1$). Hence $x_2^2 \le |x_2|$, i.e. $|x_2| \le 1$, and $x_1^2 \le |x_2| - x_2^2 \le \frac{1}{4}$ (the maximum of $s - s^2$ is $\frac{1}{4}$), i.e. $|x_1| \le \frac{1}{2}$.

Then $\cos x_1 > 0$, and since $-x_2 \cos x_1 = x_1^2 + x_2^2 \geq 0$ we get $x_2 \leq 0$. So every equilibrium lies in $|x_1| \leq \frac{1}{2}$, $-1 \leq x_2 \leq 0$.

(ii) Combine the equations. Subtract (I) from (II):

$$x_1 + x_2 + x_1 \sin x_2 - x_2 \cos x_1 = 0 \quad \Longleftrightarrow \quad x_1 (1 + \sin x_2) = -x_2 (1 - \cos x_1). \qquad \text{(III)}$$

In the region from (i): $1 + \sin x_2 \geq 1 - \sin 1 \approx 0.159 > 0$, $-x_2 \geq 0$, $1 - \cos x_1 \geq 0$. So the right side is ≥ 0, hence $x_1 \geq 0$.

(iii) Rule out $x_1 > 0$. Suppose $x_1 > 0$. Using $1 - \cos x_1 \leq x_1^2/2$ and $|x_2| \leq 1$ in (III): $x_1(1 + \sin x_2) \leq x_1^2/2$, so $1 + \sin x_2 \leq x_1/2 \leq \frac{1}{4}$, which forces $\sin x_2 \leq -\frac{3}{4}$, i.e. $x_2 \leq \arcsin(-0.75) \approx -0.848$. Feeding this back into $x_1^2 \leq |x_2|(1 - |x_2|)$ (from (i)) gives $x_1^2 \leq 0.848 \cdot 0.152 \approx 0.129$, so $x_1 \leq 0.359$; then $1 + \sin x_2 \leq 0.180$, $\sin x_2 \leq -0.820$, $x_2 \leq -0.961$, so $x_1^2 \leq 0.961 \cdot 0.039 \approx 0.037$, $x_1 \leq 0.194$; then $1 + \sin x_2 \leq 0.097$, i.e. $\sin x_2 \leq -0.903$. (Each time we use that $s(1 - s)$ decreases for $s > \frac{1}{2}$, so its largest value on $[|x_2|_{\min}, 1]$ is at the left end.) But on $[-1, 0]$, $\sin x_2 \geq \sin(-1) \approx -0.841$. Contradiction. Hence $x_1 = 0$, and we already solved that case.

$$\boxed{\text{equilibria: } x_0^{(1)} = (0, 0), \qquad x_0^{(2)} = (0, -1)}$$

(The script additionally runs a dense numerical root search on $[-3, 3]^2$ and finds exactly these two points.)

Step 2 — Jacobian (symbolic)

Differentiate term by term (Concepts §9.3 table):

$$\frac{\partial f_1}{\partial x_1} = 2x_1 - x_2 \sin x_1, \quad \frac{\partial f_1}{\partial x_2} = 2x_2 + \cos x_1, \quad \frac{\partial f_2}{\partial x_1} = 1 + 2x_1 + \sin x_2, \quad \frac{\partial f_2}{\partial x_2} = 1 + 2x_2 + x_1 \cos x_2.$$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} 2x_1 - x_2 \sin x_1 & 2x_2 + \cos x_1 \\ 1 + 2x_1 + \sin x_2 & 1 + 2x_2 + x_1 \cos x_2 \end{bmatrix}.$$

Step 3 — Evaluate at $(0, 0)$

$\sin 0 = 0$, $\cos 0 = 1$:

$$A^{(1)} = \begin{bmatrix} 0 - 0 & 0 + 1 \\ 1 + 0 + 0 & 1 + 0 + 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \boxed{\Delta \dot{x} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \Delta x, \quad \Delta x = x}$$

Eigenvalues: $\text{tr} = 1$, $\det = 0 \cdot 1 - 1 \cdot 1 = -1$, so $\lambda^2 - \lambda - 1 = 0$, $\lambda = \frac{1 \pm \sqrt{5}}{2} \approx 1.618, \ -0.618$. $\det < 0 \Rightarrow$ **saddle, unstable**.

Step 4 — Evaluate at $(0,-1)$

$\sin 0 = 0$, $\cos 0 = 1$, $\sin(-1) = -\sin 1$, $\cos(-1) = \cos 1$:

$$A^{(2)} = \begin{bmatrix} 0-(-1)\cdot 0 & -2+1 \\ 1+0-\sin 1 & 1-2+0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1-\sin 1 & -1 \end{bmatrix} \approx \begin{bmatrix} 0 & -1 \\ 0.1585 & -1 \end{bmatrix}.$$

$$\boxed{\Delta\dot{x} = \begin{bmatrix} 0 & -1 \\ 1-\sin 1 & -1 \end{bmatrix} \Delta x, \qquad \Delta x = \begin{bmatrix} x_1 \\ x_2 + 1 \end{bmatrix}}$$

Eigenvalues: $\text{tr} = -1$, $\det = 0\cdot(-1) - (-1)(1-\sin 1) = 1 - \sin 1 \approx 0.1585$:

$$\lambda = \frac{-1 \pm \sqrt{1 - 4(1-\sin 1)}}{2} = \frac{-1 \pm \sqrt{4\sin 1 - 3}}{2} \approx -0.198,\ -0.802.$$

$\text{tr} < 0$, $\det > 0$ ⇒ both negative real ⇒ **stable node, locally asymptotically stable**.

Check: `python/task2_linearization.py`; figure `task2b_stable_node.png`.

Task 2c — State space and linearization of the I/O relation

$$\ddot{y} + (3 + \dot{y}^2)\,\dot{y} + (1 + y + y^2)\,u = 0, \qquad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y \\ \dot{y} \end{bmatrix}.$$

Step 1 — Nonlinear state-space model (Concepts §7.2)

Solve for the highest derivative: $\ddot{y} = -(3+\dot{y}^2)\dot{y} - (1+y+y^2)u$. Then

$$\boxed{\begin{aligned} \dot{x}_1 &= x_2 && =: f_1(x,u) \\ \dot{x}_2 &= -(3+x_2^2)\,x_2 - (1+x_1+x_1^2)\,u && =: f_2(x,u) \\ y &= x_1 && =: g(x,u) \end{aligned}}$$

Expanded: $f_2 = -3x_2 - x_2^3 - u - x_1 u - x_1^2 u$.

Step 2 — Check that the origin is an equilibrium

$f_1(0,0) = 0$, $f_2(0,0) = -(3+0)\cdot 0 - (1+0+0)\cdot 0 = 0$ ✓. So $(x^*, u^*) = (0,0)$ is an equilibrium and $y^* = g(0,0) = 0$.

(Bonus: all equilibria.) $f_1 = 0 \Rightarrow x_2 = 0$; then $f_2 = -(1+x_1+x_1^2)u = 0$. The factor $1 + x_1 + x_1^2 = (x_1 + \frac{1}{2})^2 + \frac{3}{4} > 0$ never vanishes, so $u^* = 0$. Hence the equilibria are $(x_1^*, 0)$ with **any** x_1^* and $u^* = 0$ — a whole line. Constant nonzero inputs have no equilibrium.

Step 3 — Jacobians

$$\frac{\partial f}{\partial x} = \begin{bmatrix} 0 & 1 \\ -(1+2x_1)\,u & -3-3x_2^2 \end{bmatrix}, \qquad \frac{\partial f}{\partial u} = \begin{bmatrix} 0 \\ -(1+x_1+x_1^2) \end{bmatrix}, \qquad \frac{\partial g}{\partial x} = [1 \quad 0], \qquad \frac{\partial g}{\partial u} = 0.$$

(For $\partial f_2/\partial x_2$: $\frac{\partial}{\partial x_2}(-3x_2 - x_2^3) = -3 - 3x_2^2$. For $\partial f_2/\partial x_1$: $\frac{\partial}{\partial x_1}(-(1+x_1+x_1^2)u) = -(1+2x_1)u$.)

Step 4 — Evaluate at $(x^*, u^*) = (0, 0)$

$$\boxed{A = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0, \qquad \Delta\dot{x} = A\Delta x + B\Delta u, \;\; \Delta y = C\Delta x}$$

Here $\Delta x = x$, $\Delta u = u$, $\Delta y = y$ because the operating point is the origin. Note that $\partial f_2/\partial x_1 = -(1 + 2x_1)u$ vanishes because $u^* = 0$: around the origin the dependence of the gain on y is a second-order effect.

Equivalent linear I/O relation. $\Delta\dot{x}_1 = \Delta x_2$, $\Delta\dot{x}_2 = -3\Delta x_2 - \Delta u$, $\Delta y = \Delta x_1$ give

$$\Delta\ddot{y} + 3\,\Delta\dot{y} + \Delta u = 0, \qquad G(s) = \frac{-1}{s(s+3)}.$$

This is exactly what you get by dropping all products of small quantities ($\dot{y}^3$, yu, y^2u) in the original equation — a good sanity check.

Step 5 — Interpretation

A is upper triangular, so its eigenvalues are 0 and -3. The eigenvalue 0 reflects the line of equilibria: shifting y by a constant (with $u = 0$, $\dot{y} = 0$) gives another equilibrium, so a perturbation of y neither decays nor grows. Because one eigenvalue has zero real part, the linearization alone cannot decide the stability of the nonlinear system (Concepts §10.3).

Check: `python/task2_linearization.py` computes A, B, C, D, the eigenvalues and $G(s)$ symbolically.

Task 3 — Evolution of a discrete-time system

$$x[k+1] = \underbrace{\begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix}}_{A} x[k] + \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{B} u[k], \qquad y[k] = x[k] \;\; (C = I, \; D = 0).$$

Preparation — eigenvalues and eigenvectors of A (Concepts §12.1)

$\operatorname{tr} A = -3$, $\det A = 0 \cdot (-3) - (-2) \cdot 1 = 2$:

$$\det(\lambda I - A) = \lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2) = 0 \implies \lambda_1 = -1, \quad \lambda_2 = -2.$$

- $\lambda_1 = -1$: $A + I = \begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix}$, row gives $v_1 - 2v_2 = 0 \Rightarrow v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Check: $Av_1 = \begin{bmatrix} 0 - 2 \\ 2 - 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \end{bmatrix} = -v_1$ ✓.
- $\lambda_2 = -2$: $A + 2I = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix}$, row gives $v_1 - v_2 = 0 \Rightarrow v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Check: $Av_2 = \begin{bmatrix} -2 \\ 1 - 3 \end{bmatrix} = -2v_2$ ✓.

Notice: $B = v_2$. The input enters exactly along the eigenvector of $\lambda_2 = -2$. This will matter in (c) and (d).

3(a) — Iteration and closed form ($u \equiv 0$, $x[0] = [1,\ 0]^\top$)

Iteration ($x[k+1] = Ax[k]$, row times column):

$$x[1] = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot 1 + (-2) \cdot 0 \\ 1 \cdot 1 + (-3) \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

$$x[2] = A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 - 2 \\ 0 - 3 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}, \qquad x[3] = A \begin{bmatrix} -2 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 + 6 \\ -2 + 9 \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}.$$

Closed form, method 1 (decompose $x[0]$, Concepts §12.2). Write $x[0] = c_1 v_1 + c_2 v_2$:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \iff \begin{cases} 2c_1 + c_2 = 1 \\ c_1 + c_2 = 0 \end{cases} \iff c_1 = 1,\ c_2 = -1.$$

Since $A^k v_i = \lambda_i^k v_i$:

$$\boxed{x[k] = (-1)^k \begin{bmatrix} 2 \\ 1 \end{bmatrix} - (-2)^k \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2(-1)^k - (-2)^k \\ (-1)^k - (-2)^k \end{bmatrix}}$$

Check: $k = 0$: $[2 - 1,\ 1 - 1] = [1, 0]$ ✓; $k = 1$: $[-2 + 2,\ -1 + 2] = [0, 1]$ ✓; $k = 2$: $[2 - 4,\ 1 - 4] = [-2, -3]$ ✓; $k = 3$: $[-2 + 8,\ -1 + 8] = [6, 7]$ ✓.

Closed form, method 2 (general A^k, Concepts §12.3). $V = [v_1\ v_2] = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, $\det V = 1$, $V^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$, $\Lambda = \mathrm{diag}(-1, -2)$:

$$A^k = V \Lambda^k V^{-1} = (-1)^k \begin{bmatrix} 2 \\ 1 \end{bmatrix} [1 \quad -1] + (-2)^k \begin{bmatrix} 1 \\ 1 \end{bmatrix} [-1 \quad 2] = (-1)^k \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} + (-2)^k \begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix},$$

$$\boxed{A^k = \begin{bmatrix} 2(-1)^k - (-2)^k & -2(-1)^k + 2(-2)^k \\ (-1)^k - (-2)^k & -(-1)^k + 2(-2)^k \end{bmatrix}}$$

Check: $k = 0$ gives I; $k = 1$ gives $\begin{bmatrix} -2 + 2 & 2 - 4 \\ -1 + 2 & 1 - 4 \end{bmatrix} = A$ ✓. Multiplying by $x[0] = [1, 0]^\top$ picks the first column — the same result as method 1.

3(b) — Convergence and internal stability

For $x[0] = [1, 0]^\top$: $x[k]$ contains $-(-2)^k [1, 1]^\top$, whose size is $2^k \to \infty$ with alternating sign $(1, -2, 4, -8, \ldots)$. The other part $(-1)^k [2, 1]^\top$ stays bounded but keeps flipping sign. So $\|x[k]\| \to \infty$: **the state does not converge; it diverges with alternating sign** (as $x[3] = [6, 7]$ already suggests).

For arbitrary $x[0]$: decompose $x[0] = c_1 v_1 + c_2 v_2$ (always possible, v_1, v_2 are a basis), so $x[k] = c_1 (-1)^k v_1 + c_2 (-2)^k v_2$.

Case	Behaviour of $x[k]$
$c_2 \neq 0$ (i.e. $x[0]$ not a multiple of $[2,1]^\top$)	diverges ($\lvert\lambda_2\rvert = 2 > 1$)
$c_2 = 0$, $c_1 \neq 0$ ($x[0] = c_1[2,1]^\top$)	bounded but **not convergent**: alternates between $c_1[2,1]^\top$ and $-c_1[2,1]^\top$ ($\lambda_1 = -1$ on the unit circle)
$x[0] = 0$	stays 0

Hence $x[k]$ **converges only for** $x[0] = 0$. In the language of Concepts §13.2: the spectral radius is $\max|\lambda_i| = 2 > 1$, so the system is **not internally stable (unstable)**. Even without λ_2, the eigenvalue $\lambda_1 = -1$ alone would prevent asymptotic stability.

3(c) — Closed form with input

From Concepts §11.3:

$$x[k] = A^k x[0] + \sum_{i=0}^{k-1} A^{k-1-i} B\, u[i], \qquad k \geq 1.$$

Because $B = v_2$ is an eigenvector with $\lambda_2 = -2$, $A^j B = (-2)^j B$ (Concepts §12.5), so the sum simplifies dramatically:

$$\boxed{x[k] = A^k x[0] + \Big(\sum_{i=0}^{k-1} (-2)^{k-1-i}\, u[i]\Big) \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad A^k \text{ as in 3(a)}}$$

and $y[k] = x[k]$. *Check $k = 1$:* $x[1] = Ax[0] + (-2)^0 u[0]B = Ax[0] + Bu[0]$ ✓. *Check $k = 2$:* $A^2x[0] + (-2u[0] + u[1])B$, and directly $x[2] = Ax[1] + Bu[1] = A^2x[0] + ABu[0] + Bu[1] = A^2x[0] - 2Bu[0] + Bu[1]$ ✓.

For $x[0] = [1,0]^\top$ this reads componentwise:

$$x[k] = \begin{bmatrix} 2(-1)^k - (-2)^k + s_k \\ (-1)^k - (-2)^k + s_k \end{bmatrix}, \qquad s_k = \sum_{i=0}^{k-1} (-2)^{k-1-i} u[i].$$

Interpretation: the input only ever excites the $\lambda = -2$ mode; the $\lambda = -1$ mode is untouched by u (the difference $x_1 - x_2 = (-1)^k(x_1[0] - x_2[0])$ evolves independently of the input).

3(d) — Impulse response and BIBO stability

With $x[0] = 0$ and $u[k] = \delta[k]$, only the $i = 0$ term of the sum survives:

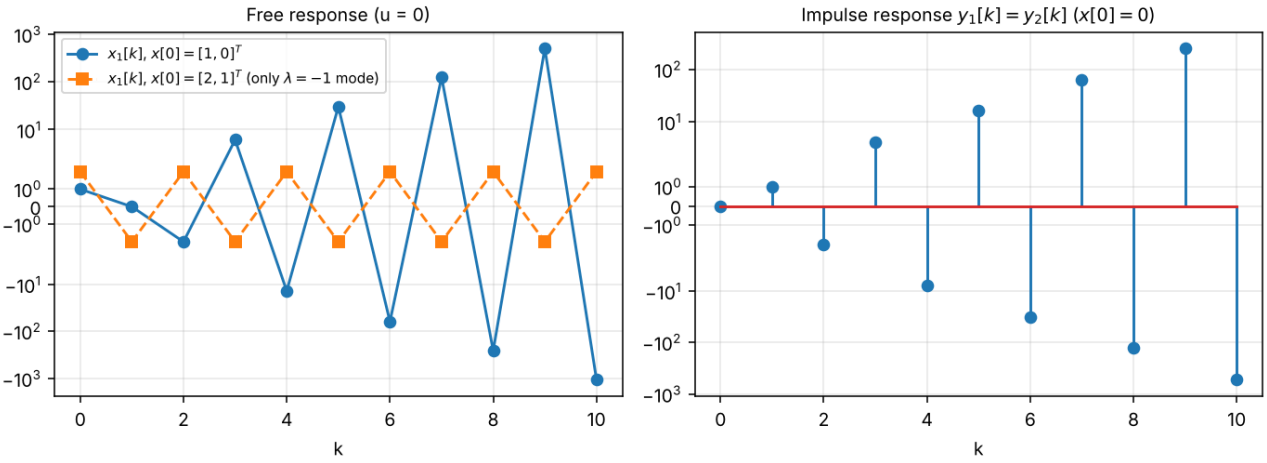
$$x[0] = 0, \qquad x[k] = A^{k-1}B = (-2)^{k-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (k \geq 1), \qquad y[k] = x[k].$$

Values: $y[1] = [1,1]$, $y[2] = [-2,-2]$, $y[3] = [4,4]$, $y[4] = [-8,-8]$, ..., $|y_i[k]| = 2^{k-1} \to \infty$.

The input $\delta[k]$ is bounded ($|u[k]| \leq 1$), but the output is unbounded. **The output does not remain bounded ⇒ the system is not BIBO stable.**

Link to (b). The unstable eigenvalue $\lambda_2 = -2$ is excited directly by the input (since $B = v_2$) and seen by the output ($C = I$), so internal instability shows up in the I/O behaviour. If B had been along v_1 instead, the impulse response would be $(-1)^{k-1} v_1$ — bounded, though not decaying — even though the system is still internally unstable. This is why internal stability and BIBO stability are different concepts (Concepts §13.3).

Check: `python/task3_discrete_time.py` verifies all numbers, both closed forms, the forced response against a simulation with random inputs, and plots the responses:



Summary of results

- **Task 1:** a) N / Y / Y; b) N / N / Y; c) Y / N / N; d) Y / N / Y (causal / TI / linear).
- **Task 2a:** equilibria $(k\pi, (-1)^k)$; $A = \text{diag}(-1, -1)$ for even k (stable), $A = \text{diag}(1, -1)$ for odd k (saddle).
- **Task 2b:** equilibria $(0, 0)$ and $(0, -1)$ only; $A^{(1)} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ (saddle, $\lambda = \frac{1 \pm \sqrt{5}}{2}$), $A^{(2)} = \begin{bmatrix} 0 & -1 \\ 1 - \sin 1 & -1 \end{bmatrix}$ (stable node, $\lambda \approx -0.198, -0.802$).
- **Task 2c:** $\dot{x}_1 = x_2$, $\dot{x}_2 = -(3 + x_2^2)x_2 - (1 + x_1 + x_1^2)u$, $y = x_1$; at the origin $A = \begin{bmatrix} 0 & 1 \\ 0 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$, $C = [1\ 0]$, $D = 0$, i.e. $\Delta\ddot{y} + 3\Delta\dot{y} = -\Delta u$.
- **Task 3:** $x[1] = [0, 1]$, $x[2] = [-2, -3]$, $x[3] = [6, 7]$; $x[k] = (-1)^k[2, 1] - (-2)^k[1, 1]$; diverges, converges only for $x[0] = 0$ (eigenvalues $-1, -2$); $x[k] = A^k x[0] + \sum_{i<k} (-2)^{k-1-i} u[i]\,[1, 1]$; impulse response $(-2)^{k-1}[1, 1]$ is unbounded ⇒ not BIBO stable.